

INNOVATIONS IN FINANCE

LECTURE 3 : THE RISE OF SMART FINANCE BLOCKCHAIN, TOKENIZATION & WEB3

Make an impact



INTRODUCTION



Make an impact



HANDS UP

No judgement in this room today — but let's be precise about what we own

1 Who owns crypto?

Most hands stay up.

2 Keep it up if you can say what the protocol actually does.

Some come down.

3 Keep it up if you have read anything the project itself published.

More come down.

4 Keep it up if you would still hold it knowing the price will never move again.

Almost none left.

This is not a criticism of you. It is a description of how the asset was sold to you.

You own an exposure. That is a legitimate thing to own — but it is not the same thing as understanding what you hold. By the end of today you will.





WHAT DID YOU ACTUALLY BUY?

The question your other finance courses answer for every asset except this one

A share

A claim on future cash flows.

You can value it: discount the cash flows. That is what your corporate finance course is about.

A bond

A contractual stream of payments.

You can value it: the payments are written down and dated.

A flat

A right to occupy, or to collect rent.

You can value it: comparable rents, comparable sales.

Bitcoin

...?

No cash flow. No issuer. No claim on anything. Its value comes entirely from what the next buyer will pay.

THIS IS NOT A MORAL JUDGEMENT

Gold has worked exactly this way for five thousand years, and nobody calls it a scam.

BUT IT HAS A CONSEQUENCE

None of your valuation tools apply. Anyone giving you a price target for bitcoin is doing something other than valuation.





NOW SEPARATE TWO THINGS

This is the move that makes the rest of the session possible

The asset

Bitcoin, Ether, and several thousand others.

Volatile, speculative, no cash flow, priced by opinion. Interesting as a market phenomenon. This is the part you already know about — and it is the part that gets all the attention.

The infrastructure

A shared register that several parties who do not trust each other can update without a central operator, and that settles instantly.

Boring. Invisible. This is the part that is changing finance, and it is what the rest of today is about.

THE ANALOGY THAT MAKES IT OBVIOUS

You can think email is trivial and still recognise that TCP/IP changed the world. The 2010s taught everyone to look at the coins. The interesting object was always underneath them.

For the next ninety minutes we ignore prices completely.

Not because they do not matter to you — because they are the least interesting thing here. We come back to your portfolio at the end.





PREVIOUSLY ON *INNOVATIONS IN FINANCE*

◆ **Recap:**

- AI → smarter decision-making
- APIs → financial connectivity
- Embedded finance → invisible banking



LEARNING OBJECTIVES

◆ Knowledge (know/understand):

- Understand the fundamentals of blockchain, smart contracts, and tokenization.
- Identify how DeFi and Web3 are challenging traditional financial infrastructure..

◆ Skills (be able to):

- Compare centralized vs. decentralized models.
- Analyze real-world use cases of blockchain in finance.
- Evaluate the potential and limits of token-based finance.

➔Curiosity, creativity and critical thinking will be rewarded



WHERE WE ARE GOING

One question to keep in your pocket the whole session

THE TEST WE APPLY TO EVERYTHING TODAY

Does this solve a problem somebody would pay to have solved?

1 Why it exists

The settlement problem: what actually happens between buying a share and owning it.

2 How it works

Built with your own hands, in this room, with croissants and push-ups.

3 Digital money

Stablecoins, CBDCs, and the vocabulary mess everyone gets wrong.

4 Tokenization

The Eiffel Tower, BlackRock, and a French money market fund.

5 DeFi

A bank with glass walls — where you can read the mechanics of finance.

6 Your portfolio

We come back to the board, and you do the analysis.





WHY DOES ANY OF THIS EXIST?

Start with a problem, not with a technology

You buy a share on Monday. You are not the owner until Wednesday. What happens in between?

Monday	The trade	You and a seller agree a price. That takes microseconds.
Monday → Wednesday	The plumbing	Broker, clearing house, custodians and registrars reconcile who owes what to whom. Different systems, different databases, no shared view.
In the meantime	The cost	Collateral is posted and frozen. Somebody carries counterparty risk. Whole teams exist to reconcile the two sides.
Wednesday	Settlement	The register is finally updated. You own the share.

THIS IS A LIVE PROBLEM, NOT A MUSEUM PIECE

The US moved to T+1 in May 2024. The EU and the UK are targeting October 2027. Shortening it by one day is a multi-year, continent-wide project.

THE BLOCKCHAIN PROPOSITION

One shared register that all parties update together. Nothing to reconcile, because there is only one copy. That is the whole pitch.





THE FOUR QUESTIONS

Apply these to any blockchain project — including the ones on the board

1

Are there several parties who do not fully trust each other?

If there is only one party, you have an internal record-keeping problem, not a blockchain problem.

2

Do they need to share the same register?

Not just exchange messages — actually agree on one state of the world, at the same time.

3

Is the absence of a central operator an advantage, or just a constraint you are accepting?

Sometimes a trusted operator is exactly what everybody wants. Sometimes no party will accept another one holding the pen.

4

Would an ordinary database do this better?

Faster, cheaper, editable, and it can forget things. This is the question nobody asks, and it is where most projects die.

Four yeses and you have a real case. Most projects of the last decade could not answer question 4.





BLOCKCHAIN FUNDAMENTALS



Make an impact

LET'S BUILD A BLOCKCHAIN TOGETHER!

The Blockchain Café ☕🍰

 **Customers** – make transactions (“I buy a croissant for 1 coin”)

 **Ledger Keepers (Nodes)** – record every transaction on their sheet

 **Auditor / Miner** – checks if all ledgers match before approving a block

 **Hacker (optional)** – tries to alter a transaction afterward

“In blockchain, no single person is trusted — everyone *collectively* maintains the truth.”



LET'S BUILD A BLOCKCHAIN TOGETHER!

The Blockchain Café ☕🍰





WHAT JUST HAPPENED AT THE BLOCKCHAIN CAFÉ?

What You Did

Each node wrote the same transaction

Auditor checked all copies before approval

Hacker couldn't change one record alone

We didn't need a central waiter

What It Represents in Blockchain

Distributed ledger – everyone holds a full copy

Consensus mechanism

Immutability & security

Decentralization – trust the network, not a single entity

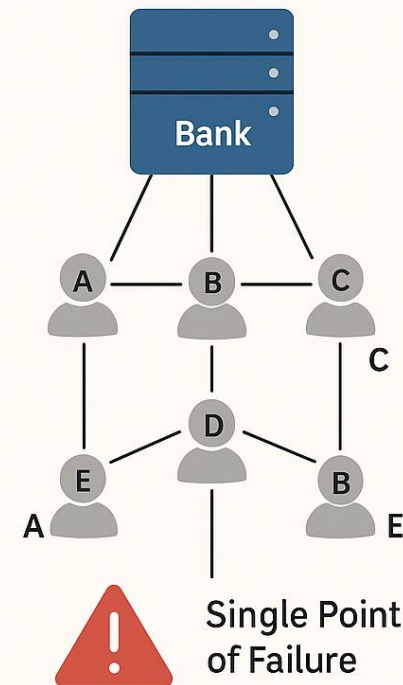


WHAT IS BLOCKCHAIN?

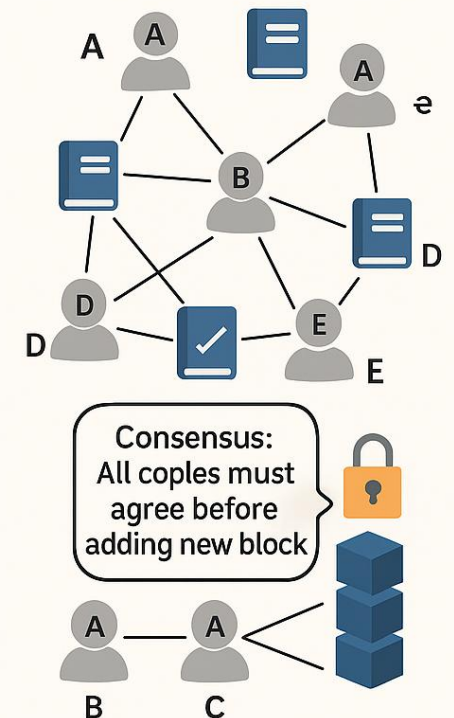
- ◆ Blockchain is a shared, digital ledger that records transactions in a secure, transparent, and decentralized way.
- ◆ Instead of being stored in a single central server (like a traditional bank), blockchain data is copied and stored across thousands of computers around the world.
- ◆ Once a block of data is added, it can't be changed—that's what makes it immutable.
- ◆ Before a new transaction is recorded, computers on the network must agree (reach consensus) that it's valid.
 -  **Decentralized**
 -  **Transparent**
 -  **Tamper-proof**

What is Blockchain?

Centralized Ledger



Decentralized Ledger





HOW TO REACH A CONSENSUS ?

Consensus Type	How It Works	Pros	Cons
Proof of Work	Compete through effort	Secure, proven	Energy-intensive
Proof of Stake	Chance based on what you risk	Energy-efficient, faster	Favors the wealthy (“the rich get to validate”)

HOW TO REACH A CONSENSUS ?

- ◆ Consensus Mechanism = Trust Without a Middleman
- ◆ 2 Main algorithms :

Proof of Work (PoW)

🔄 How it works: Miners compete to solve complex mathematical puzzles

✅ Pros: Highly secure, well-tested

⚠️ Cons: Energy-intensive, slow



Bitcoin

<https://btcscan.org/>

Proof of Stake (PoS)

🔄 How it works: Validators are selected based on the amount of crypto they "stake"

✅ Pros: Energy-efficient, faster

⚠️ Cons: Risk of centralization, "rich get richer" effect



Ethereum (since The Merge)

<https://etherscan.io/>



SMART CONTRACTS: PROGRAMMABLE FINANCE

◆ What is a Smart Contract?

- Self-executing code stored on a blockchain
- Automatically enforces terms of an agreement
- No intermediary, no manual execution

◆ Key Features:

- Trustless (runs without human intervention)
- Transparent (code is public)
- Immutable (can't be altered once deployed)

Use Case: Automated Loan Disbursement

Traditional system

- Bank verifies loan request
- Manual approval → delays
- Disbursement subject to business hours, intermediaries

With Smart Contract

- Conditions coded (e.g. credit score > X, collateral present)
- Auto-disbursement triggered instantly
- 24/7 execution, no manual check, fewer costs

SELF-CUSTODY AND WALLETS

- ◆ Crypto Assets are working with three pieces of information:
 - The address associated with a balance and used for sending and receiving funds
 - The address' corresponding public key
 - The address' corresponding private key
- ◆ The creation of a blockchain account starts with the generation of a private key (the corresponding public key is derived using an asymmetric cryptography mechanism)
- ◆ This private key represents the ownership of the account and is unique
- ◆ In blockchain mechanism, the user can store this key by himself (and all the associated responsibilities)

« Not your keys,
not your coins »





WHAT YOU SHOULD NOT PUT ON A BLOCKCHAIN

The slide that will make you the most useful person in any project meeting

AN ORDINARY DATABASE WINS ON

Speed	Thousands of writes per second, not tens.
Cost	Orders of magnitude cheaper to run.
Simplicity	Any developer knows how. Few know smart contracts.
Correction	You can fix a mistake. On-chain, a mistake is permanent.
Privacy	Nothing is public unless you decide it is.

THE CONFLICT NOBODY MENTIONS

A blockchain is immutable by design. Nothing can ever be erased.

The GDPR gives every European a right to erasure.

These two statements cannot both be satisfied for the same personal data. It is not a bug to be patched — it is a direct contradiction between a technical property and a legal right.

HOW SERIOUS PROJECTS HANDLE IT

Put the proof on-chain, and the data off-chain. A hash — a fingerprint — proves a document existed and has not changed, without the document itself ever touching the register. Delete the document, and the fingerprint becomes meaningless. This is exactly how the digital identity model on the next slide works.

"Why not just use a database?" is the most valuable question you can ask in a blockchain meeting. Ask it early.





DIGITAL IDENTITY MANAGEMENT

◆ What is Self-Sovereign Identity (SSI)?

- A decentralized approach to identity where users own and control their credentials
- Blockchain stores credentials' proofs, not the personal data itself
- User presents only the required claims, not full documents (selective disclosure)

How It Works in Financial Services (KYC/AML/Onboarding)

1. Credential Issuance

- A trusted issuer (e.g., a bank, gov't) verifies your ID once
- Issues a **verifiable credential**, digitally signed and linked to your wallet

2. Selective Disclosure in KYC/AML

- When applying for a loan or opening an account, you **share only what's needed** (e.g., "over 18", "EU citizen")
- The verifier (e.g., fintech, bank) checks the validity via the blockchain without storing the data

3. Credit Assessment & Reusability

- Credit score, salary history, or transaction records can be stored as credentials
- These are portable across institutions and **verified instantly**

Privacy by Design

- ✓ No central storage of identity = No honeypots for hackers
- ✓ Only **cryptographic proofs** on-chain (hashes of data)
- ✓ ID layers ensure **granular consent** for each use case

Focus: Europe's EUDI Wallet

EU The European Digital Identity Wallet (EUDI) aims to:

- Allow EU citizens to **store identity, diplomas, medical data** in a unified wallet
- Be compliant with **eIDAS 2.0 regulation**
- Enable frictionless access to public and private services (finance, telecom, gov.)



STABLECOINS & CBDCS



Make an impact



CRYPTO-ASSET ≠ CRYPTOCURRENCY

Clean the vocabulary up now, or every sentence afterwards is confused

To be money, something has to do three jobs. Score bitcoin on each.

Medium of exchange

Can you pay with it?

Almost nowhere, and the fee and confirmation time make small payments impractical.

No

Unit of account

Are prices quoted in it?

Nothing is priced in BTC. Even crypto companies quote in dollars — including their own salaries.

No

Store of value

Does it hold purchasing power?

Over ten years, spectacularly. Over any given quarter, it can lose half.

It depends

SO USE THE RIGHT WORDS

Crypto-asset

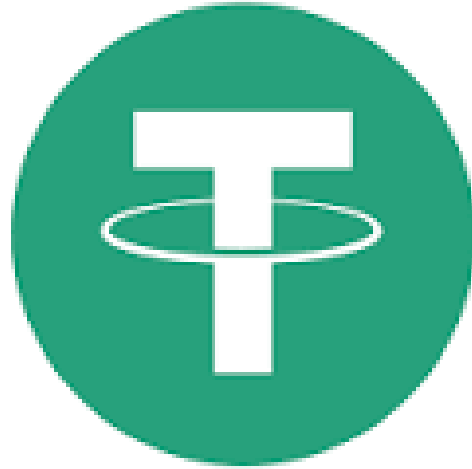
an investment instrument recorded on a blockchain. Bitcoin, Ether, and most of the board.

Cryptocurrency

a crypto-asset that actually functions as money. Stablecoins are the only serious candidates.



THREE TYPES OF DIGITAL MONEY





HOW STABLECOINS WORK

- ◆ **Collateralized stablecoins:** backed 1:1 by fiat or bonds (USDC, USDT)
- ◆ **Crypto-collateralized:** backed by overcollateralized crypto (DAI)
- ◆ **Algorithmic:** use code + market incentives (Terra/LUNA... until it broke 🌟)
- ◆ Transparency and audits determine **trustworthiness**.



ALGORITHMS CAN FAIL...





THE STABLECOIN USE CASE THAT IS NOT ABOUT CRYPTO

Forget trading. Look at a company paying a supplier on another continent.

TODAY, VIA CORRESPONDENT BANKS

- 2 to 5 days for most corridors
- Two to four intermediaries, each taking a fee
- An FX spread you do not see
- No visibility on where the money is
- Nothing moves at the weekend

SETTLED IN A STABLECOIN

- Minutes, not days
- One transfer, no correspondent chain
- The conversion cost is explicit
- Both sides see the same transaction
- Saturday works exactly like Tuesday

SO WHY HAS EVERY COMPANY NOT SWITCHED?

Accounting

How does your auditor treat it, and how is it presented in the accounts?

Counterparty

"Backed 1:1" is a claim. Backed by what, held where, audited by whom?

The edges

Getting in and out of the banking system is still the slow, expensive part.

Since MiCA, a euro stablecoin issued in Europe is a regulated product with reserve and redemption rules. That is what makes a treasurer able to consider it.



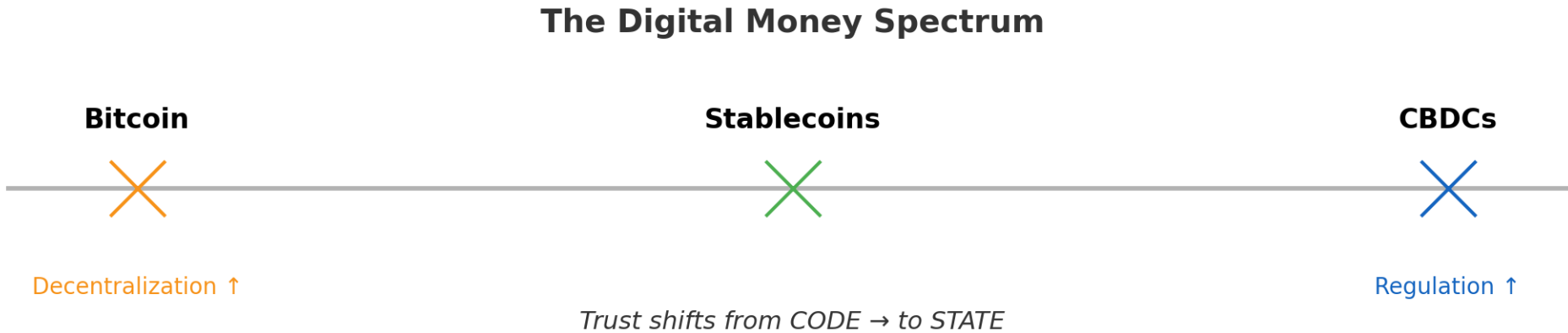


ENTER THE CBDCS

When Central Banks Join the Game

- ◆ CBDC = Central Bank Digital Currency
- ◆ Issued directly by a central bank, legal tender, digital form of fiat.
- ◆ Can be retail (for citizens) or wholesale (for banks).
- ◆ Goals:
 - improve payment efficiency,
 - fight illicit use,
 - preserve monetary sovereignty.
- ◆ Examples: China's e-CNY, ECB's Digital Euro, Bahamas' Sand Dollar.

STABLECOINS VS CBDCs: WHO DO YOU TRUST?



- ◆ All digital, but not all equally decentralized
- ◆ Blockchain reshapes trust models:
- ◆ from “trust banks” → “trust networks” → “trust governments again (digitally)”.
- ◆ The key question: **Who controls the money — code, market, or state?**



TOKENIZATION IN FINANCE



Make an impact

CASE: OWNING A SLICE OF THE EIFFEL TOWER?

- ◆ Didactic Example: Tokenizing the Eiffel Tower
 1. Valuation: Eiffel Tower hypothetical market value = €10B
 2. Legal Structuring: SPV (Special Purpose Vehicle) created to hold rights
 3. Token Creation: 10 million tokens issued → 1 token = €1,000 worth
 4. Regulatory Compliance: KYC/AML onboarding, regulated exchange listing
 5. Distribution: Investors purchase tokens via a digital platform
 6. Secondary Market: Tokens traded peer-to-peer or on exchanges
 7. Revenue Sharing: Token holders receive share of ticket sales or rental income
 - ✓ Democratized access
 - ✓ Increased liquidity
 - ✓ Transparent ownership
 - ✓ Programmable income via smart contracts





WHAT IS TOKENIZATION?

Turning Real Assets into Digital Tokens

- ◆ What is Tokenization?
 - The process of converting rights to a real-world asset into a digital token on a blockchain.
 - Each token represents ownership or a claim on a share of an asset (e.g. real estate, bonds, art).
- ◆ Use Cases:
 - Real Estate: Fractional ownership of buildings or land
 - Private Equity & Bonds: Liquidity in traditionally illiquid assets
 - Commodities: Trade gold, oil, or carbon credits via tokens
 - Luxury Goods & Art: Ownership and provenance tracking
 - NFTs: Unique tokens for digital/physical assets (e.g. art, collectibles, IP rights)
- ◆ Benefits:
 - Global access: Investors worldwide can participate
 - Fractional ownership: Democratizes access to large assets
 - Faster settlement: 24/7 markets with instant transactions
 - Transparent & auditable: Built-in tracking and compliance
- ◆ Challenges:
 - Regulatory uncertainty (e.g., securities classification)
 - Need for reliable asset valuation and custody
 - Technical integration with traditional systems

NFTS IN LUXURY GOODS: THE ARIANEE X BREITLING CASE

◆ Brand Objective:

- Enhance customer relationship through a secure, digital-first ownership experience
- Guarantee authenticity and product traceability
- Empower customers with control over personal data and privacy

◆ NFT Integration with Arianee:

- Every Breitling watch now includes a blockchain-based digital passport (NFT)
- NFT contains details like:
 - ◆ Proof of ownership
 - ◆ Product origin (materials, sourcing, etc.)
 - ◆ Warranty and repair history
 - ◆ Acts as a digital twin of the physical watch

◆ Customer Benefits:

- Authenticity check without third parties
- Track repairs, extend warranties, and manage insurance
- Maintain anonymous ownership and full control over personal data
- Easy resale with verified provenance (secondary market trust)

◆ Benefits for Luxury Brands:

- NFTs as tools for loyalty, ownership verification, and after-sales service
- Unlock new digital customer experiences and secondary market control
- Promotes circular economy and product lifecycle traceability



.arianee



INVESTMENT MANAGEMENT: A NEW PARADIGM

- ◆ Traditional Model – Mutual Funds Today
 - Requires custodians, transfer agents, and clearing systems
 - T+2 settlement with risk of settlement failure
 - Manual, fragmented, and costly processes
 - Limited real-time transparency for investors



BlackRock

- ◆ Tokenized Mutual Funds – Blockchain Model:
 - Smart contract-based fund shares issued as tokens
 - Real-time Delivery vs Payment (DvP) on blockchain
 - Removes intermediaries → faster, cheaper, programmable
 - Greater investor access (24/7, fractional ownership, global reach)
- ◆ BlackRock USD Institutional Digital Liquidity Fund
 - Mainframe asset management (invest in US sovereign debt, mainly treasuries)
 - Fully tokenized



A VERY BORING PROBLEM

A French mid-sized company is sitting on €2m of cash. Where does it go?

OPTION A — THE CURRENT ACCOUNT

Available instantly. Earns essentially nothing.

On €2m, over a year, that is a five-figure sum of interest simply not collected.

OPTION B — A MONEY MARKET FUND

- Cut-off at midday — miss it and you wait
- Settlement the next day, sometimes later
- Minimum subscription amounts
- Nothing happens at the weekend

SO THE TREASURER'S REAL PROBLEM

Cash that might be needed on Thursday cannot be committed on Monday — because getting it back takes too long and the calendar does not cooperate.

So it stays in the current account earning nothing. Not because the treasurer is careless, but because the operational friction is worse than the yield.

The problem is not the product. The problem is the settlement cycle.

Nobody in this room finds that exciting. Hold on to that feeling.

It is the least glamorous problem in corporate finance, it affects hundreds of thousands of European companies, and it is worth real money.

Run the four questions on it. We answer number four in two slides.





SPIKO: WHAT IS ACTUALLY ON THE BLOCKCHAIN

Not the money. The shareholder register.

This is the sentence to take out of today: tokenizing a fund does not turn it into a crypto-asset. It replaces the registrar.

WHAT IT IS

- A UCITS money market fund, approved by the AMF
- Invested in short-dated Treasury bills — euro and dollar versions
- Spiko is regulated as a MiFID investment firm by the ACPR and the AMF
- Founded in 2023 by two former senior civil servants from financial regulation

WHAT THE BLOCKCHAIN CHANGES

- The register of unit-holders lives on-chain, not at a transfer agent
- Subscribe and redeem continuously — no midday cut-off, no weekend gap
- Small tickets become economic, because the admin cost collapses
- Interest accrues daily and is visible in the token

ANSWERING QUESTION 4: WHY NOT A DATABASE?

Because the register has to be trusted and read by parties who do not answer to each other — the manager, the depositary, the auditor, the corporate holding the units, and increasingly the platforms that accept those units as collateral. A database owned by one of them is a database the others must take on faith. That is exactly the case the technology was built for — and it is why this works where most projects do not.





WHY THIS IS THE BEST CASE IN THE COURSE

The most boring instrument in finance, on the most hyped technology

THE JUXTAPOSITION IS THE ARGUMENT

A money market fund holding government T-bills is as far from speculation as finance gets. Zero glamour. The risk-free rate, and nothing else.

If the technology only worked for gambling, it would not be here.

AND THE INSTITUTIONS ARRIVED

BlackRock launched a tokenized treasury fund in 2024.

Spiko and Amundi — Europe's largest asset manager — launched SAFO, a tokenized UCITS fund built for corporate treasury and collateral needs.

WHAT THIS TELLS YOU, AND IT IS NOT WHAT YOU EXPECT

The technology did not win where it was loudest. It is winning in fund administration, collateral management and settlement — the parts of finance no one writes threads about. Meanwhile the applications that got all the attention for a decade have mostly not survived contact with a regulator or a balance sheet.

Nobody will ever tell you a story about a tokenized money market fund at a party.

It is also the thing on this slide most likely to be part of your job.





WHERE MY COMPANY SITS IN THIS: KOWIKAN × SPIKO

Disclosure again — Kowikan is my company and Spiko is a partner. Same rules as last time.

1

Forecast the cash

Kowikan builds the 30/60/90-day treasury forecast: what comes in, what goes out, and when.

2

Find what is genuinely idle

The forecast tells you how much cash is not needed before a given date. That number is the whole point.

3

Place it — and get it back

Tokenized money market fund units. Subscribe on Monday, redeem on Thursday because a client paid late. No cut-off, no weekend.

4

The treasury decision becomes continuous

Not a monthly arbitrage with a bank. An ongoing one, driven by the forecast.

ASSEMBLY AGAIN, NOT INVENTION

Kowikan does not manage a fund and does not hold anyone's money. Spiko does that, under AMF supervision. We do the forecasting layer. Two regulated specialists, one workflow — the same pattern as Voxpay in session 2.

THE OBJECTIONS I GET, AND SHOULD

"What does my auditor say?" · "Who is the depositary if something fails?" · "How is it treated in the accounts?" · [TO COMPLETE — add the one that comes up most, and your honest answer, including where you do not have one yet]

[TO COMPLETE — what we have measured so far, or say openly that we are still building and measuring]





CHALLENGES & LIMITS

Tokenization Is Powerful — but Not Magic

- ◆ **Regulatory fragmentation** (MiCA, Pilot Regime, SEC uncertainty).
- ◆ **Custody and settlement still depend on traditional infrastructure.**
- ◆ **Valuation and governance of real assets remain complex.**
- ◆ **Market adoption still small** — need trusted intermediaries.
- ◆ **Key question:** Can we tokenize everything, or only what investors understand and regulators accept?



WEB3 & DECENTRALIZED FINANCE

Make an impact



WHICH INSTITUTIONS OFFER THESE TODAY?

Deposit – Borrow – Exchange – Invest

WHAT IS DEFI?

◆ DeFi = Decentralized Finance

- A global financial system built on blockchains (mainly Ethereum)
- Replaces banks and brokers with smart contracts

◆ How It Works:

- Peer-to-peer (P2P) financial services
- Powered by self-executing smart contracts
- Non-custodial: users keep control of their funds

◆ Core Concepts:

- **Yield Farming:** Earn rewards (often tokens) by providing liquidity to DeFi protocols
- **Staking:** Lock crypto in a protocol to secure the network and earn interest
- **Liquidity Pools:** Smart contracts that hold token pairs (e.g., ETH/USDC) to facilitate swaps and earn fees
- **DEX** : Decentralized Exchange

Benefits of DeFi:

- ✓ Open 24/7, global access
- ✓ Transparent and auditable
- ✓ Composability ("Money Legos") – apps build on each other





AAVE: A SECURED LOAN WITH THE LID OFF

What a credit committee decides in private, written as numbers anyone can read

	A BANK	AAVE
The collateral	A credit file, a guarantee, a mortgage deed.	You deposit €6,000 of ETH. The contract holds it.
The loan-to-value	Set by internal policy. You never see it.	Published. Borrow up to a fixed percentage of the collateral.
The decision	A committee, a scorecard, days or weeks.	A function call. Seconds. No one is asked.
The margin call	A phone call, a letter, a negotiation.	A liquidation threshold in the code. No negotiation exists.
The enforcement	Legal proceedings, months, recovery uncertain.	Anyone may liquidate you and collect a bonus for doing it.

Why 150% collateral and not 105%?

Because the collateral is volatile and liquidating it takes time. That is exactly the reasoning behind a haircut on repo collateral — same problem, same arithmetic, different vocabulary. DeFi did not invent this. It made it visible.





EXERCISE: AT WHAT PRICE DO YOU LOSE EVERYTHING?

Five minutes. Paper, not phones. This is a margin call, and you will never forget it.

YOUR POSITION

Collateral	2 ETH, and ETH is trading at €3,000 → €6,000
You borrow	€3,600 of a euro stablecoin
Liquidation threshold	80% — if your debt exceeds 80% of the collateral value, you are liquidated
Liquidation penalty	5% of the position, paid to whoever liquidates you

FIND THREE NUMBERS

- At what ETH price are you liquidated?
- What percentage fall is that from €3,000?
- How much do you lose to the penalty?

Then: was this a good idea?

THE MECHANISM, IN ONE LINE

You are safe while $\text{debt} < \text{collateral value} \times \text{liquidation threshold}$.

So the liquidation price is the collateral value at which that stops being true — and nothing else about you matters. Not your income, not your history, not who you are.

This is a margin call. Every leveraged position in every market you will ever work in has one.





UNISWAP — THE MARKET MAKER WITHOUT A TRADER

◆ Traditional Market Maker

- Trader A wants to exchange EUR for USD :
EUR/USD for 5
- Market Maker (Bank/Broker) provides bid/ask spread : **1,1705/1,1710**
- FX Execution via interbank or dealer platform :
Mine
- Fees/Spreads embedded in price

◆ Result :

- Centralized liquidity provision
- Spreads and fees determined by intermediaries
- Slippage risk and limited transparency

◆ Uniswap AMM

- Trader A connects wallet to Uniswap
- Swap EUR/USDC via a liquidity pool (EUR/USDC pair)
- *Smart contract* calculates price using constant product formula ($x * y = k$)
- Liquidity Providers earn fees automatically

◆ Result :

- Liquidity provided by users (LPs), not intermediaries
- Price adjusts based on supply & demand
- 24/7 market access without permission
- Fees go to LPs, not banks



THE LOSS WITH THE WORST NAME IN FINANCE

Impermanent loss — what happens to a market maker who never updates their quotes

A HUMAN MARKET MAKER

Quotes a bid and an offer. Watches the market.

When news arrives, they move their quotes — fast — so nobody can trade against them at a stale price.

That reflex is the job. It is what the spread pays for.

AN AUTOMATED MARKET MAKER

Quotes from a formula: $x \times y = k$. It has no news, no view, no reflex.

When the market moves, arbitrageurs buy the cheap side off it until the pool price matches the world.

The liquidity provider is on the losing end of every one of those trades.

SO THE NAME IS WRONG IN TWO WAYS

It is not impermanent — it only reverses if the price comes back, which is a bet, not a property. And it is not really a loss to a bug: it is the cost of providing liquidity without being able to react. Fees are what compensate you for it. Sometimes they do not.

You are being paid a spread to be systematically picked off by better-informed traders. That is not a DeFi concept — it is adverse selection, and it is as old as markets.





WHERE DOES THE YIELD COME FROM?

The single most useful question in finance — and DeFi is the only place you can fully answer it

Trading fees

Real

Someone paid to swap. You provided the liquidity that let them. This is genuine revenue from a genuine service.

Borrowing interest

Real

Someone borrowed your capital and pays for it. Also genuine — and it is what a bank does.

Token emissions

Not yield

The protocol prints new governance tokens and gives them to you. Your return is paid in dilution of a thing whose price depends on more people arriving.

Someone else's deposit

Run the other way

If the return is funded by new entrants rather than by activity, you have found the oldest structure in finance and it has a name.

Any yield you cannot decompose into these boxes is one you do not understand yet. That rule outlives DeFi.





DAOS AND FINANCIAL GOVERNANCE

From Shareholders to Token Holders

- ◆ **A DAO (Decentralized Autonomous Organization)** is a digital-native organization run by smart contracts and governed collectively by its members (no central authority).
 - Rules encoded on the blockchain
 - Transparent, automatic execution
 - Decisions made through token-based voting
- ◆ **Governance:**
 - **Members vote** on proposals using governance tokens
 - Every token = 1 vote (typically)
 - Used to decide: new features, partnerships, budgets, treasury usage
 - Examples:
 - ◆ **Uniswap DAO** votes on fee structures
 - ◆ **Aave DAO** manages risk parameters and lending protocols



WHAT DEFI PROVED, AND WHAT IT DID NOT

Be honest about both — the failure is more interesting than the success

IT GENUINELY PROVED

- **Lending can be automated**

Collateral, rates, margin calls, liquidation — with no human in the loop, ever.

- **Market making can be automated**

A formula and a pool replaced a dealer, and it works.

- **Settlement can be instant**

No T+2. The trade and the settlement are the same event.

- **Composability is real**

Protocols plug into each other without asking permission. Nothing in traditional finance does this.

IT DID NOT SOLVE

- **Ordinary humans**

Seed phrases, gas fees, irreversible mistakes. The onboarding is brutal and has barely improved.

- **Not exploding**

Bridge hacks, oracle manipulation, contract bugs. Billions lost, no recourse, no deposit insurance.

- **Credit without collateral**

This is the big one. See below.

- **Being anyone's main financial system**

After a decade, it remains overwhelmingly used to trade crypto with crypto.

Lending to someone who already has 150% of the money is not credit. It is a pawnshop.

The hard part of banking — deciding who to trust when there is no collateral — is precisely the part DeFi has not touched. It rebuilt the operations layer beautifully and never reached the credit layer. Same conclusion as session one, from a different direction.





RISKS, REGULATION, AND TRUST



Make an impact



WHAT CAN GO WRONG?

Innovation Runs Faster Than Risk Control

- ◆ **Smart-contract bugs** → irreversible loss (DAO Hack 2016 – \$60 m)
- ◆ **Protocol hacks / bridge exploits** → billions stolen (Ronin, Poly Network...)
- ◆ **Stablecoin failures** → loss of peg (Terra/LUNA 2022 – \$40 bn)
- ◆ **Custody risk** → lost private keys, exchange bankruptcies (FTX)
- ◆ **Systemic risk** → interconnected DeFi protocols amplify contagion



THE REGULATORY RESPONSE

◆ Europe:

- MiCA (Markets in Crypto Assets)
 - ◆ Adopted in 2023, in force from 2024–2025
 - ◆ Rules on: Stablecoins
 - ◆ Licensing for crypto service providers
 - ◆ Market abuse & investor protection

◆ USA: SEC & Ongoing Legal Debate

- SEC actions against Binance, Coinbase, Ripple (XRP)
- Dispute on whether tokens are securities
- Lack of clear classification is causing uncertainty



INSTITUTIONAL BRIDGES: REGULATED DEFI

Can Code and Compliance Coexist?

- ◆ **“Regulated DeFi” platforms: KYC wallets, permissioned pools.**
- ◆ **Project Mariana (BIS + SNB + MAS): FX swaps on tokenized deposits.**
- ◆ **Aave Arc, JP Morgan Onyx, SDX → institutional DeFi pilots.**
- ◆ **Goal = keep efficiency of smart contracts + maintain AML/CFT controls.**



THE FUTURE OF TRUST

From Institutions to Infrastructure — and Back Again.

- ◆ **Blockchain = trust in network**
- ◆ **Smart contracts = trust in code**
- ◆ **Regulation = trust in rules**
- ◆ **The future is hybrid finance (TradFi × DeFi)**
- ◆ **Goal: combine efficiency of code + protection of law**



SEVENTEEN YEARS IN: THE HONEST SCORECARD

Bitcoin's whitepaper is from 2008. Enough time has passed to judge.

SURVIVED CONTACT WITH REALITY

- **Stablecoins as settlement rails**

Real volumes, real corporate use, now regulated in Europe under MiCA.

- **Tokenized funds and collateral**

BlackRock, Amundi, Spiko. Boring, growing, institutional.

- **Product traceability**

Breitling and Arianee. A digital passport for a physical object, with a real business case.

- **Instant settlement as a benchmark**

Even institutions that will never touch a public chain now ask why settlement takes two days.

- **Bitcoin as a store of value**

Contested, volatile — but it is seventeen years old and it did not disappear.

DID NOT

- **Bitcoin as a payment method**

The founding promise. Seventeen years later, essentially nobody pays with it.

- **NFTs as an art market**

A speculative bubble that deflated almost entirely. The technology survived; that use of it did not.

- **DAOs as a way to run things**

Voter apathy, token concentration, legal limbo. One token one vote is plutocracy, not democracy.

- **Consumer DeFi**

Still unusable for anyone who does not enjoy the difficulty.

- **Algorithmic stablecoins**

Terra/LUNA erased around \$40bn in 2022. The idea itself was flawed, not the execution.

Everything on the left is infrastructure. Everything on the right was a consumer product or a token.

That is not a coincidence, and it is the answer to the question we started with.





BACK TO THE BOARD

Your holdings. Your grid. Your analysis — not mine.

Take one thing written on that board
— ideally one you own —
and run the four questions on it.

- 1 Several parties who do not trust each other?
- 2 Do they need a shared register?
- 3 Is having no central operator an advantage here?
- 4 Would an ordinary database do this better?

HOW IT RUNS

4 min in pairs, written

4 min three pairs report

Rule no defending, no attacking — just answer the four questions honestly

AND THEN THE HONEST PART

Some of what is on that board will score four out of four. Some will score zero. Owning something that scores zero is not stupid — plenty of people own gold, and gold scores zero on every framework ever built.

But you are now able to tell the difference between owning an asset and understanding an infrastructure. That was the whole point of today.





INTRO – BEFORE WE TALK BLOCKCHAIN...

Your wallet $\neq$ your portfolio strategy



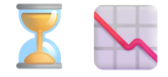
- ◆ You might own BTC, ETH, DOGE... or a meme that went to zero.
- ◆ But do you manage it, or just hope?
- ◆ **Finance still obeys the same laws: risk, return, and time horizon.**

DIVERSIFICATION 101: DON'T MARRY YOUR COIN 💔

Put your eggs in different baskets (and watch the baskets!)

- ◆ **Diversification** = spreading exposure across uncorrelated assets.
- ◆ Reduces portfolio volatility without necessarily reducing returns.
- ◆ **Crypto + equities + bonds + cash → build a near zero correlation**
- ◆ Correlation in stress = 1 😬.

VOLATILITY, TIME, AND THE DREADED MAX DRAWDOWN



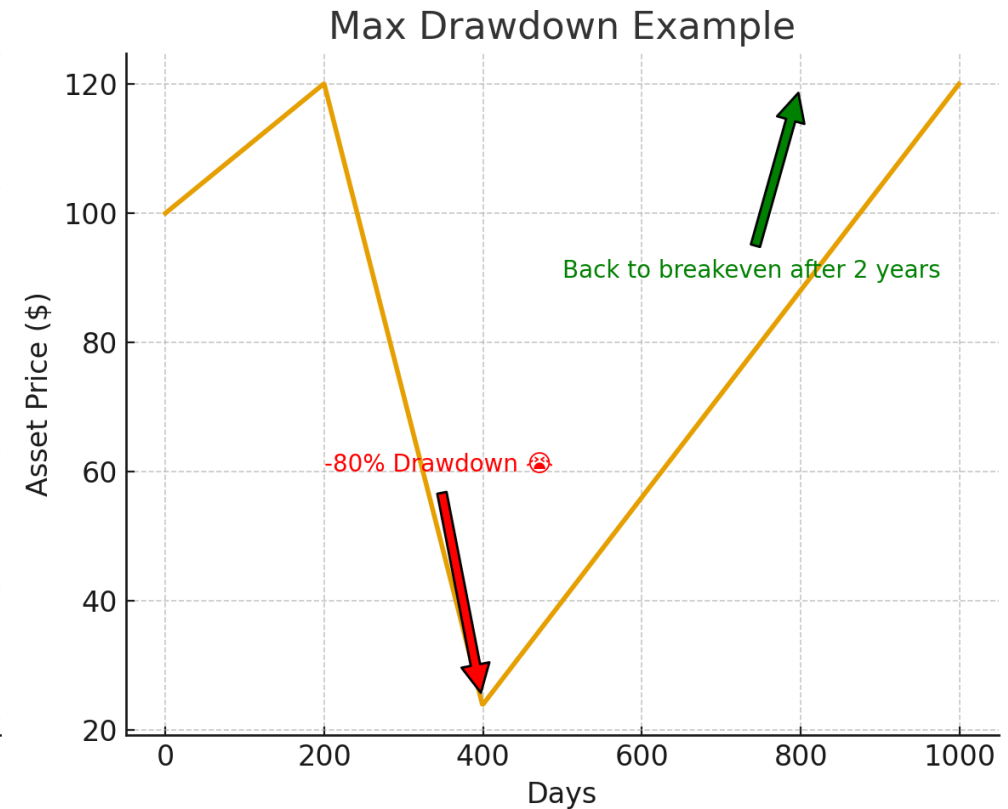
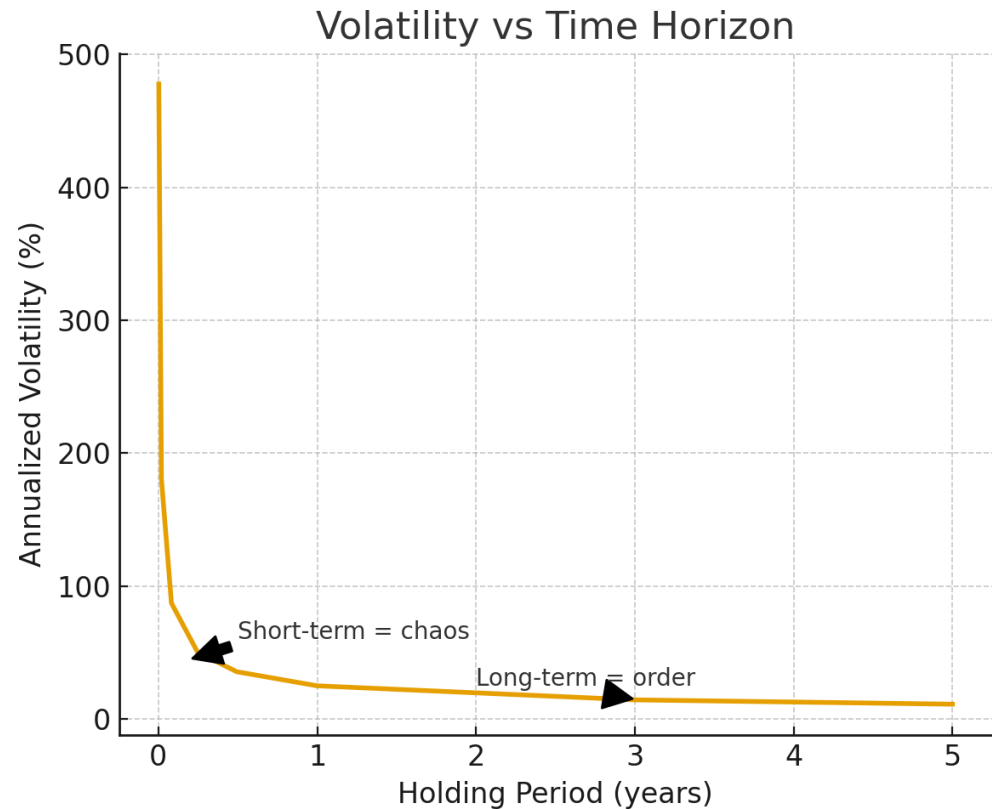
The longer you invest, the more you can survive storms

- ◆ **Volatility** ↓ as holding period ↑ (time smooths returns).
- ◆ Short-term = price noise; long-term = fundamentals.
- ◆ **Max Drawdown: largest peak-to-trough loss before recovery.**
- ◆ Example: “My coin dropped –80% but I’m still HODLing” = max drawdown champion 🏆.



WHAT IT LOOKS LIKE ON A CHART

Time heals (most) volatility — until it doesn't!





WRAP-UP



Make an impact



WHAT TO TAKE OUT OF TODAY

Five things — the first one is the whole session

1

The blockchain is not an asset. It is a register that settles.

Bitcoin is one application of it, and a speculative one. Tokenized funds are another, and a boring one. Only the second is changing finance.

2

Always ask question four.

Would an ordinary database do this better? Faster, cheaper, editable, and it can forget — which European law requires. Most projects die on this question.

3

Tokenizing means replacing the registrar, not creating a coin.

Spiko put the fund's shareholder register on-chain. The fund is still a UCITS money market fund approved by the AMF.

4

DeFi rebuilt the operations of finance and never reached the credit.

Lending against 150% collateral is a pawnshop. Deciding who to trust without collateral is the hard part, and it is untouched.

5

Decompose every yield you are ever offered.

Fees and interest are real. Newly printed tokens are dilution. Money from new entrants is something else entirely.

Before next week: find one tokenization project announced in the last year and score it out of four. Bring the score and one line of justification.





BLOCKCHAIN, TOKENIZATION & WEB3 ESSENTIALS

What is the main purpose of a smart contract in finance?

- A. To replace central banks
- B. To enable automatic and trustless execution of agreements
- C. To issue cryptocurrencies
- D. To store personal identity data on-chain





BLOCKCHAIN, TOKENIZATION & WEB3 ESSENTIALS

Which of the following is an example of a stablecoin that failed due to algorithmic flaws?

- A. USDC
- B. TerraUSD (UST)
- C. Tether (USDT)
- D. Binance USD (BUSD)



BLOCKCHAIN, TOKENIZATION & WEB3 ESSENTIALS

What is the formula used by Uniswap's automated market maker (AMM) model?

- A. $x + y = z$
- B. $x - y = k$
- C. $x * y = k$
- D. $x^2 + y^2 = z^2$



BLOCKCHAIN, TOKENIZATION & WEB3 ESSENTIALS

True or False: Tokenization allows fractional ownership of real-world assets like real estate or mutual funds.



BLOCKCHAIN, TOKENIZATION & WEB3 ESSENTIALS

What is one major challenge users face with Web3 wallets?

- A. Too many customer service agents
- B. Over-regulation
- C. Losing access due to lack of password recovery
- D. Instant transaction approvals





Make an impact

